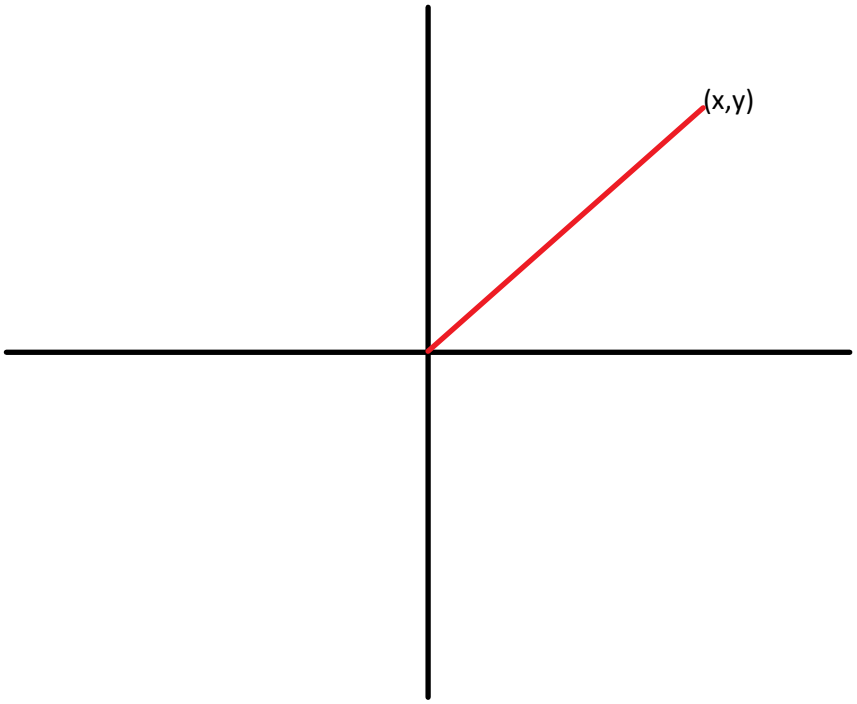
Section 9.7 Other Coordinate Systems

You might recall from Calculus 1 or Trigonometry Polar Coordinates…….



So we can get to any point in the x-y plane using this coordinate system. Turns out there is something similar for points in 3 dimensions We call is cylindrical coordinates.

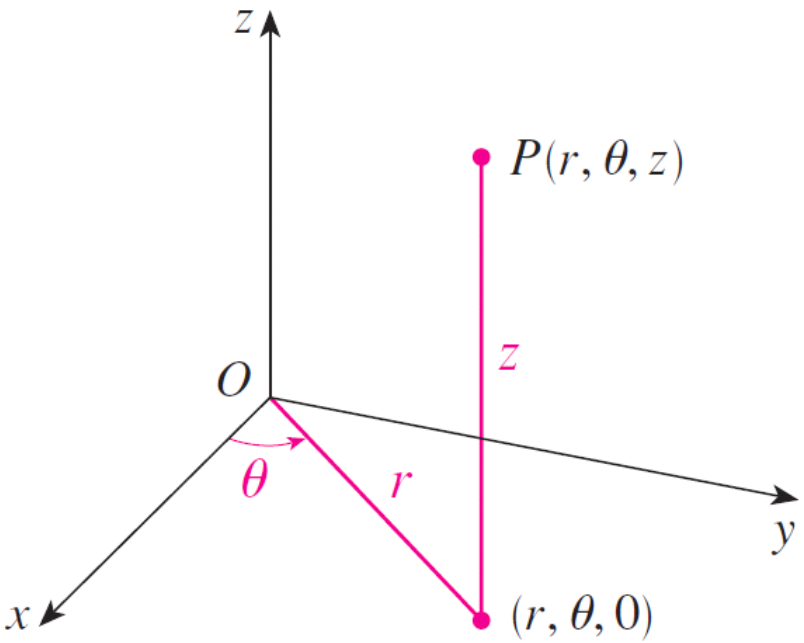


FIGURE 2

The cylindrical coordinates of a point

Section 9.7 Other Coordinate Systems

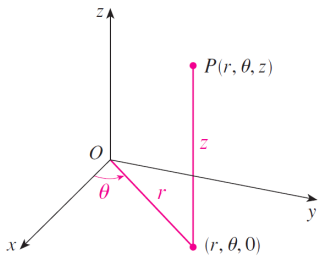


FIGURE 2
The cylindrical coordinates of a point

1

$$x = r\cos\theta \qquad y = r\sin\theta \qquad z = z$$

2

$$r^2 = x^2 + y^2 \qquad \tan\theta = \frac{y}{x} \qquad z = z$$

Hw Example Plot the points

3. (a) $(2, \pi/4, 1)$ (b) $(4, -\pi/3, 5)$

Hw Example Convert to Cylindrical coordinates

5. (a) $(1, -1, 4)$ (b) $\left(-1, -\sqrt{3}, 2\right)$

Section 9.7 Other Coordinate Systems

Spherical coordinates: Instead of using z to get our height we can use an angle measured from the x-y plane….

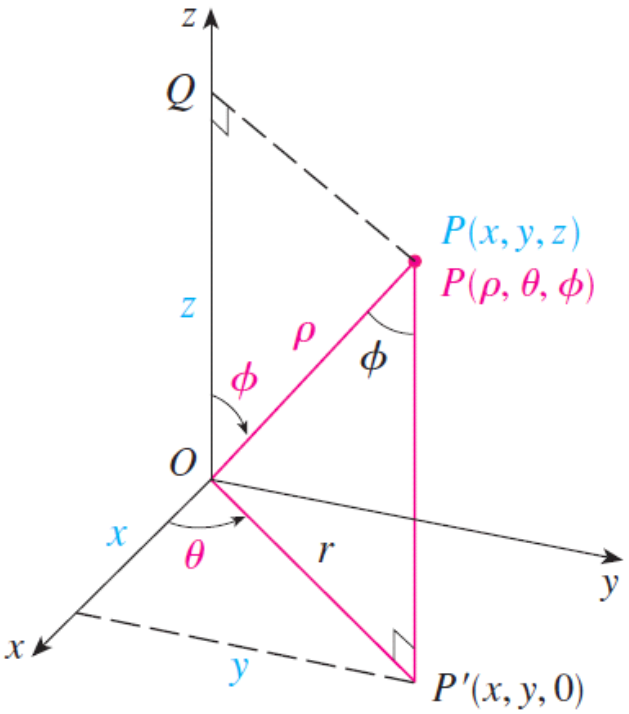


FIGURE 10

3 $$x = \rho \sin\phi \cos\theta \qquad y = \rho \sin\phi \sin\theta \qquad z = \rho \cos\phi$$

4 $$\rho^2 = x^2 + y^2 + z^2$$

Section 9.7 Other Coordinate Systems

Hw Example Plot the points whose spherical coordinates are given and then convert them to rectangular coordinates.

3 $x = \rho \sin\phi \cos\theta \qquad y = \rho \sin\phi \sin\theta \qquad z = \rho \cos\phi$

4 $\rho^2 = x^2 + y^2 + z^2$

7. (a) $(1, 0, 0)$ (b) $(2, \pi/3, \pi/4)$

Section 9.7 Other Coordinate Systems

Surfaces in cylindrical and spherical coordinates.

HW Example Identify the graph of $z = 4 - r^2$

Section 9.7 Other Coordinate Systems

Converting equations

Convert

21. $x^2 + y^2 = 2y$

To cylindrical and spherical coordinates.